

ABSTRACT

The Digital RPM Meter Using Arduino is an efficient and accurate electronic system designed to measure the rotational speed of motors and rotating objects in terms of Revolutions Per Minute (RPM). The project is based on an Arduino Nano microcontroller, which acts as the central processing unit of the system. An IR sensor module is used to detect the rotation of the shaft or rotating object by sensing the reflection of infrared light from a reflective strip attached to the rotating surface. Each time the reflective strip passes in front of the sensor, a pulse signal is generated and sent to the Arduino.

The project consists of major components including Arduino Nano, IR sensor module, 16×2 LCD display with I2C module, LED indicator, resistor, breadboard, connecting wires, USB power supply, and a DC motor or rotating shaft for testing. The IR sensor continuously monitors the rotating object and sends digital pulse signals to the Arduino. The Arduino counts these pulses within a fixed time interval and calculates the RPM mathematically. The calculated RPM value is then displayed in real time on the 16×2 LCD display.

An LED indicator is also included in the circuit to indicate pulse detection and proper system operation. The resistor is used for current limiting and protection of the LED. The I2C module simplifies LCD interfacing by reducing wiring complexity and minimizing the number of Arduino pins required.

The system provides high accuracy, fast response time, compact design, low power consumption, and reliable performance. The project demonstrates the practical implementation of sensors, embedded systems, microcontroller programming, and digital measurement techniques. It is highly useful in industrial automation, robotics, motor speed monitoring, laboratory experiments, and educational applications where accurate rotational speed measurement is required.

TABLE OF CONTENTS

.Abstract.....	i
CHAPTER 1 INTRODUCTION	1
1.1 Introduction to Digital RPM Meter	11
1.2 Arduino Nano	12
1.3 IR Sensor Module	13
1.4 16×2 LCD Display	14
1.4.1. I2C Module	14
1.5 DC Motor with Fan, Switch, and Battery	15
CHAPTER 2. THEORY	16
CHAPTER 3. CIRCUIT DIAGRAM	19
CHAPTER 4 DERIVIATION AND SOURCE CODE	20
CHAPTER 5COMPONENTS USED IN CIRCUIT	23
5.1 Arduino Nano	23
5.2 IR Sensor Module	24
5.3 16×2 LCD Display with I2C Module	25
5.4 DC Motor with Fan, Switch, and Battery	26
5.5 Jumper Wires and PCB	27
5.6 Power Adapter for 5V Supply	27
CHAPTER 7 ADVANTAGES	28
CHAPTER 8 DISADVATAGES	29
CHAPTER 9 APPLICATIONS	30
CHAPTER 10 CONCLUSION	31
REFRENCES	32

LIST OF FIGURES

Fig No	Description of Figure	Page No
Fig 1.2.1	Arduino Nano	12
Fig 1.3.1	IR sensor	13
Fig 1.4.1	16*2 LCD display	14
Fig 1.4.2	I2C Module	14
Fig 1.5.1	DC Motor with Fan, Switch, and	15
Fig 1.6.1	BatteryLED	15
Fig 3.1	CIRCUIT DIAGRAM	19
Fig 3.2	Actual project	19
Fig 5.1.1	Arduino Nano	23
Fig 5.2.1	IR sensor	24
Fig 5.3.1	16*2 LCD display with I2C module	25
Fig 5.4.1	DC motar with switch and fan	26
Fig 5.5.1	PCB and Jumper wires	27
Fig 5.6.1	Power Adapter	27

LIST OF TABLE

Table No	Description of Figure	Page No
Table A	List of symbols	10
Table B	List of Abbrevations	9
Table C	List of figures	8
Table 1.2.1	power pins	12
Table 1.2.2	digital pins	13
Table 1.2.3	analog pins	13

~~LIST OF ABBREVIATIONS~~

Abbreviation	Full Form
RPM	Revolutions Per Minute
IR	Infrared
LCD	Liquid Crystal Display
I2C	Inter-Integrated Circuit
DC	Direct Current
LED	Light Emitting Diode
USB	Universal Serial Bus
SDA	Serial Data Line
SCL	Serial Clock Line
VCC	Voltage Common Collector
GND	Ground
PCB	Printed Circuit Board
IDE	Integrated Development Environment
TX	Transmit
RX	Receive
IC	Integrated Circuit
ATmega	Advanced Technology Mega Microcontroller
PWM	Pulse Width Modulation

~~LIST OF SYMBOLS~~

Symbol	Meaning
V	Voltage
I	Current
R	Resistance
RPM	Revolutions Per Minute
IR	Infrared Signal
Hz	Frequency (Hertz)
s	Time in Seconds
LED	Light Emitting Diode
LCD	Liquid Crystal Display
DC	Direct Current
GND	Ground Terminal
VCC	Positive Supply Voltage
Ω	Ohm (Unit of Resistance)
A	Ampere (Unit of Current)
mA	Milliampere
W	Watt (Unit of Power)
SDA	Serial Data Line
SCL	Serial Clock Line
+	Positive Terminal
-	Negative Terminal

CHAPTER 1 : INTRODUCTION

1.1 Introduction to Digital RPM Meter

In modern industries, automation systems, robotics, and motor-control applications, measuring the rotational speed of machines and motors is extremely important. The rotational speed of a motor or rotating object is generally measured in terms of RPM (Revolutions Per Minute), which indicates how many complete rotations are made in one minute. Accurate RPM measurement helps in maintaining efficiency, performance, safety, and proper control of machines.

Traditional analog tachometers are often less accurate, difficult to read, and unable to provide continuous digital monitoring. To overcome these limitations, digital RPM meters are widely used because they provide precise, reliable, and real-time speed measurement. Digital systems are faster, easier to monitor, compact in size, and more suitable for modern electronic and industrial applications.

This project presents the design and implementation of a “Digital RPM Meter Using Arduino”, which measures the rotational speed of a rotating object using an IR sensor and Arduino Nano microcontroller. The IR sensor detects every rotation of the shaft by sensing reflected infrared light from a reflective strip attached to the rotating surface. Each detected rotation generates a pulse signal that is sent to the Arduino Nano for processing.

The Arduino continuously counts the incoming pulses and calculates the RPM value using programmed mathematical calculations. The measured RPM is then displayed on a 16×2 LCD display through an I2C communication module, which reduces wiring complexity and improves system simplicity.

The project consists of important electronic components such as Arduino Nano, IR sensor module, 16×2 LCD display, I2C module, LED indicator, resistor, breadboard, connecting wires, and regulated 5V power supply. Each component plays an important role in ensuring accurate sensing, fast processing, stable operation, and clear output display.

The system provides several advantages such as high accuracy, real-time monitoring, compact design, low power consumption, easy interfacing, and cost-effective implementation. Due to these features, the project can be widely used in industries, educational laboratories, robotics systems, motor testing applications, automation systems, conveyor monitoring, and research projects.

This project also helps in understanding the practical implementation of embedded systems, sensor interfacing, microcontroller programming, digital signal processing, and electronic instrumentation techniques. Overall, the Digital RPM Meter using Arduino is an efficient and reliable system for accurate speed measurement applications.

1.2 Arduino Nano

ArduinoNano is the main controlling unit of the Digital RPM Meter project. It is a compact microcontroller development board based on the ATmega328P microcontroller and is widely used in embedded systems, automation projects, robotics, sensor interfacing, and digital control applications. Due to its small size, low power consumption, easy programming, and high reliability, Arduino Nano is highly suitable for portable and real-time electronic systems.

In this project, the Arduino Nano acts as the brain of the entire system. It receives pulse signals from the IR sensor whenever the rotating object completes a revolution. The Arduino processes these pulse signals, performs mathematical RPM calculations, and displays the calculated rotational speed on the 16×2 LCD display. It continuously monitors the sensor output and updates the RPM value in real time, ensuring accurate and stable measurement.

The Arduino Nano is programmed using the Arduino IDE software through a USB connection. The programming language used is based on Embedded C/C++. The microcontroller executes the instructions stored in its memory and controls all operations of the RPM meter system automatically.

Main Features of Arduino Nano

- Based on ATmega328P microcontroller
- Operates at 5V DC supply
- Compact and lightweight design
- 14 digital input/output pins
- 6 analog input pins
- Built-in USB programming support
- Easy interfacing with sensors and displays
- Low power consumption
- Fast processing capability
- Suitable for embedded and automation projects

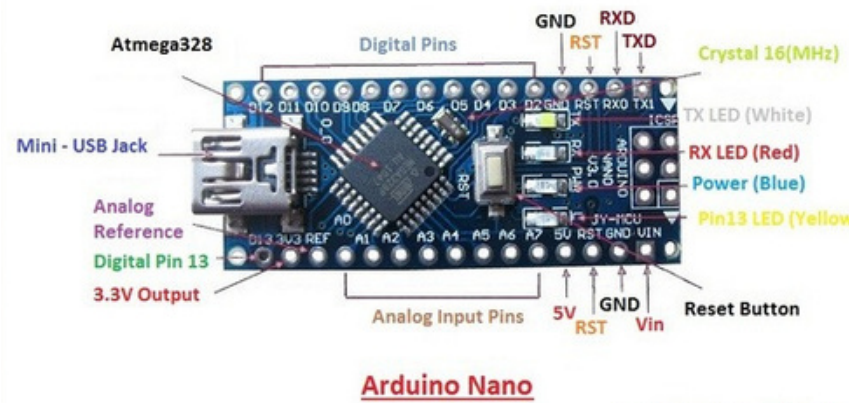


fig-1.2.1

Role of Arduino Nano in This Project

- Receives pulse signals from IR sensor
- Counts the number of rotations
- Calculates RPM using mathematical formulas
- Displays RPM value on LCD display
- Controls LED indication
- Provides real-time monitoring and processing

The Arduino Nano simplifies the overall circuit design and improves the accuracy, flexibility, and reliability of the DigitalRPM Meter

Power Pins:-

Pin	Function
VIN	External
5V	Regulate
3.3V	3.3V
GND	Ground
RESET	Resets

table 1.2.1

DigitalInput/Output Pins

Pin	Function
D0 (RX)	Serial data receive
D1 (TX)	Serial data transmit
D2	External interrupt
D3	PWM and interrupt
D4	Digital input/output
D5	PWM output pin
D6	PWM output pin
D7	Digital input/output
D8	Digital input/output
D9	PWM output pin
D10	PWM/SS pin
D11	PWM/MOSI pin
D12	MISO pin
D13	Built-in LED pin

Table 1.2.2

Analog Pins

Pin	Function
A0	Analog input
A1	Analog input
A2	Analog input
A3	Analog input
A4	SDA (I2C
A5	SCL (I2C
A6	Analog input
A7	Analog input

Table 1.2.3

1.3IRSensor Module

TheIR (Infrared) Sensor Module is used to detect the rotation of the motor or rotating object in the Digital RPM Meter project. It works on the principle of infrared light reflection. The sensor contains an IR transmitter and an IR receiver.

The IR transmitter continuously emits infrared light toward the rotating object. A reflective strip is attached to the rotating surface. Whenever the strip passes in front of the sensor, the infrared light reflects back to the receiver. The receiver detects this reflected light and generates a pulse signal.

In this project, each pulse represents one complete rotation. These pulse signals are sent to the Arduino Nano, which counts the pulses and calculates the RPM value. The calculated RPM is then displayed on the LCD display.

Features of IR Sensor Module

- Non-contact speed detection
- Fast and accurate response
- Low power consumption
- Easy interfacing with Arduino
- Compact and reliable design

Role of IR Sensor in This Project

- Detects rotation of the shaft
- Generates pulse signals
- Sends pulses to Arduino Nano
- Helps in accurate RPM calculation

The IR sensor makes the RPM meter accurate, reliable, and easy to use for real-time speed measurement applications.

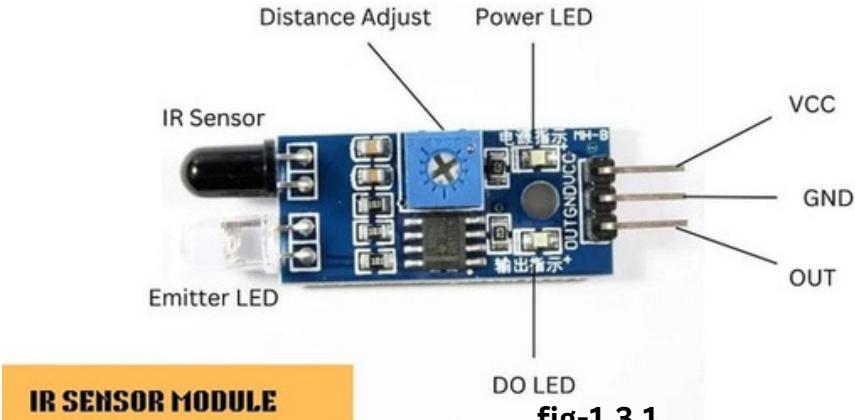


fig-1.3.1

1.4 16×2 LCD Display

A 16×2 LCD Display is used in this project to show the measured RPM value. It is an electronic display module capable of displaying 16 characters in 2 rows. The LCD helps the user monitor the rotational speed clearly and in real time.

In this project, the LCD is connected to the Arduino Nano through an I2C module, which reduces wiring complexity and uses fewer Arduino pins. The Arduino sends the calculated RPM value to the LCD, and the display continuously updates the speed reading.

The LCD provides a simple and user-friendly way to observe the output of the RPM meter system.

Features of 16×2 LCD Display

Displays RPM value clearly

- Real-time output monitoring
- Low power consumption
- Easy interfacing with Arduino
- Compact and reliable display unit

Role of LCD in This Project

- Displays calculated RPM value
- Shows real-time motor speed
- Improves user interaction
- Makes monitoring easy and accurate

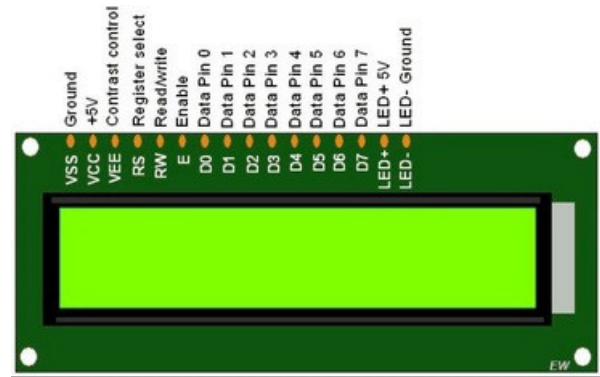


fig-1.4.1

The 16×2 LCD display improves the usability of the Digital RPM Meter by providing clear and continuous speed indication.

1.4.1 I2C Module

The I2C (Inter-Integrated Circuit) module is used to connect the 16×2 LCD display with the Arduino Nano using only two communication wires. Normally, an LCD requires many connecting wires, but by using the I2C module, the circuit becomes simpler, cleaner, and easier to manage.

The I2C module is attached at the back side of the LCD display and allows serial communication between the Arduino and LCD. It uses two main communication lines:

SDA (Serial Data Line)

SCL (Serial Clock Line)

In this project, the Arduino sends RPM data to the LCD through the I2C module, and the LCD displays the RPM value in real time.

Features of I2C Module

- Reduces wiring complexity
- Uses only two communication wires
- Saves Arduino pins
- Provides simple and fast communication
- Easy interfacing with LCD display

Role of I2C Module in This Project

- Connects Arduino Nano with LCD display
- Transfers RPM data from Arduino to LCD
- Simplifies the overall circuit design
- Makes the system compact and reliable



fig-1.4.2

The I2C module makes LCD interfacing easier and improves the overall efficiency and neatness of the Digital RPM Meter circuit.

1.5 DC Motor with Fan, Switch, and Battery

A DC motor with a fan is used in this project as the rotating object whose speed is measured by the Digital RPM Meter. The motor converts electrical energy into mechanical rotation, and the attached fan helps provide visible rotation for RPM measurement and testing.

A battery is used as the power source to supply DC voltage to the motor, while a switch is connected to control the ON and OFF operation of the circuit. When the switch is turned ON, current flows from the battery to the motor, causing the fan to rotate. The rotating fan is then detected by the IR sensor for RPM calculation.

Components Used

- DC Motor
- Fan Blade
- ON/OFF Switch
- 9V Battery
- **Role in This Project**
 - DC motor provides rotational motion
 - Fan helps in visible rotation detection
 - Battery supplies power to the motor
 - Switch controls motor operation
 - Rotating fan is used for RPM measurement

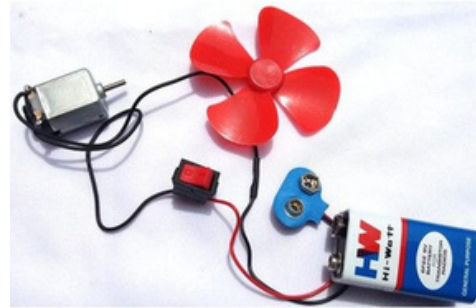


fig-1.5.1

The DC motor, fan, switch, and battery together form the practical rotating setup of the Digital RPM Meter system and help in accurate speed measurement.

1.6 LED Indicator

An LED (Light Emitting Diode) is used in the Digital RPM Meter project as an indicator to show the working status of the system. The LED glows or blinks whenever the IR sensor detects the rotation of the motor fan. The LED is connected with a resistor to limit the current and protect it from damage. It provides a simple visual indication that helps the user know whether the sensor and circuit are working properly.

Features of LED

Low power consumption

Fast response time

Bright visual indication

Compact and reliable component

Long operational life

Role of LED in This Project

- Indicates pulse detection
- Shows proper system operation
- Helps during testing and troubleshooting
- Provides visual monitoring of sensor activity



fig-1.6.1

The LED improves the usability and reliability of the Digital RPM Meter by giving clear visual feedback during operation.

CHAPTER 2 – THEORY

The Digital RPM Meter Using Arduino works on the principle of pulse detection and pulse counting. RPM stands for Revolutions Per Minute, which indicates the number of complete rotations made by a rotating object in one minute. In this project, the rotational speed of a DC motor fan is measured using an IR sensor and Arduino Nano microcontroller.

The IR sensor detects the rotation of the fan by sensing reflected infrared light from the rotating blade or reflective strip attached to it. Whenever the reflective surface passes in front of the sensor, the IR sensor generates a pulse signal. These pulse signals are sent to the Arduino Nano for processing.

The Arduino Nano continuously counts the number of pulses generated by the sensor within a fixed time interval. Using programmed mathematical calculations, the Arduino calculates the RPM value accurately. The calculated RPM is then displayed on the 16×2 LCD display through the I2C module.

The project mainly consists of Arduino Nano, IR sensor module, 16×2 LCD display with I2C module, DC motor with fan, LED indicator, resistor, switch, battery, and connecting wires. Each component performs an important function in the speed measurement process.

The system provides real-time RPM monitoring, high accuracy, fast response, low power consumption, and reliable performance. Since the measurement is performed without physical contact, the system experiences less mechanical wear and provides better operational life.

This project demonstrates the practical use of embedded systems, sensor interfacing, digital electronics, and microcontroller programming for industrial and educational applications.

2.1 Working Principle

The Digital RPM Meter works on the principle of infrared signal detection, pulse counting, and mathematical RPM calculation using Arduino Nano.

In this project, a DC motor with a fan is used as the rotating object. then gives the 5v power supply to the system An IR sensor is placed in front of the rotating fan blade or reflective strip attached to it. The IR sensor contains two main parts: an IR transmitter and an IR receiver.

The IR transmitter continuously emits infrared rays toward the rotating fan. Whenever the reflective strip or bright surface comes in front of the sensor during rotation, the emitted infrared light reflects back to the IR receiver. The receiver detects this reflected light and generates a digital pulse signal.

Each pulse generated by the sensor represents one complete rotation of the fan blade. These pulse signals are sent to the Arduino Nano microcontroller through the input pin.

The Arduino continuously reads and counts the incoming pulses using programmed coding. Inside the program, a mathematical formula is used to calculate the RPM value based on the number of pulses detected within a fixed time interval.

The RPM is calculated using:

$$RPM = Pulses\ per\ Second \times 60$$

After calculation, the Arduino sends the RPM data to the 16×2 LCD display through the I2C module. The LCD then displays the motor speed in real time.

At the same time, the LED indicator blinks whenever a pulse is detected, showing that the sensor and system are working properly.

Thus, the complete system detects rotational movement, converts it into digital pulse signals, processes the signals using Arduino programming and mathematical calculations, and finally displays the accurate RPM value on the LCD screen.

2.2 Pulse Detection by IR Sensor

Pulse detection is the main process used to measure RPM in this project. The IR sensor continuously monitors the rotating fan blade or reflective strip attached to the motor shaft. The sensor contains an IR transmitter and IR receiver. The IR transmitter emits infrared light toward the rotating object. Whenever the reflective strip passes in front of the sensor, the infrared light reflects back to the receiver. The receiver detects this reflected light and generates a digital pulse signal. These pulses are sent to the Arduino Nano for RPM calculation.

Each pulse represents one complete rotation of the motor fan. The Arduino counts these pulses within a fixed time interval and calculates the rotational speed accurately.

Pulse Detection Process

- IR transmitter emits infrared rays
- Reflective strip reflects IR light
- IR receiver detects reflected signal
- Sensor generates digital pulse
- Arduino counts the pulses
- RPM value is calculated and displayed on LCD

Thus, the IR sensor converts rotational movement into digital pulse signals, which helps the Arduino measure and display accurate RPM values in real time.

2.3 Arduino Processing and RPM Calculation

Arduino Nano acts as the main processing unit of the Digital RPM Meter system. The IR sensor module has three main pins: VCC, GND, and OUT. The OUT pin of the IR sensor is connected to the D2 digital input pin of the Arduino Nano.

When the rotating fan blade or reflective strip passes in front of the IR sensor, the sensor detects the reflected infrared light and generates a digital pulse signal from its OUT pin. This pulse signal is sent directly to the D2 pin of the Arduino Nano.

The Arduino continuously reads these incoming pulse signals through the D2 input pin. In the coding, a counter variable is used to count how many pulses are received within a fixed time interval, usually one second.

After counting the pulses, the Arduino processes the data using a mathematical formula to calculate the RPM value.

The RPM is calculated using:

$$RPM = \text{Pulses per Second} \times 60$$

For example, if 25 pulses are detected in one second:

$$RPM = 25 \times 60 = 1500$$

After calculation, the Arduino sends the RPM value to the 16×2 LCD display through the I2C module. The LCD then displays the motor speed continuously in real time.

Signal Processing Steps

- IR sensor detects rotating fan blade
- Pulse signal comes from OUT pin of sensor
- Signal is received at D2 pin of Arduino Nano
- Arduino counts the pulses using coding
- RPM formula is applied
- Calculated RPM is displayed on LCD

Thus, the Arduino converts sensor pulse signals into accurate digital RPM readings and displays them clearly on the screen.

2.4 LCD Display with I2C Communication

A 16×2 LCD display is used in the Digital RPM Meter project to show the calculated RPM value in real time. The LCD is connected to the Arduino Nano using an I2C module, which makes the circuit simpler and reduces wiring complexity.

The I2C module uses only two communication pins:

- SDA (Serial Data Line)
- SCL (Serial Clock Line)

In Arduino Nano:

- SDA pin is connected to A4
- SCL pin is connected to A5

After the Arduino calculates the RPM value from the sensor pulses, the processed data is sent to the LCD through the I2C module. The LCD continuously updates and displays the motor speed on the screen.

The I2C communication system helps in fast and reliable data transfer while using fewer Arduino pins.

Working Process

- Arduino calculates RPM value
- Data is sent through SDA and SCL lines
- I2C module receives the data
- LCD displays RPM value in real time

Advantages of I2C LCD

- Uses fewer connecting wires
- Saves Arduino pins
- Easy interfacing and coding

Thus, the LCD with I2C module provides clear and continuous RPM display, making the Digital RPM Meter

2.5 Complete Working of the Project

The Digital RPM Meter starts working when power is supplied to the circuit through the battery or USB supply. The Arduino Nano initializes the IR sensor and 16×2 LCD display.

The DC motor rotates the fan blade, and the IR sensor continuously monitors the rotating surface. Whenever the reflective strip or fan blade passes in front of the sensor, the IR sensor detects the reflected infrared light and generates a digital pulse signal from its OUT pin.

This pulse signal is sent to the D2 pin of the Arduino Nano. The Arduino continuously reads and counts these pulses using programmed coding. After counting the pulses within a fixed time interval, the Arduino applies the RPM formula and calculates the motor speed accurately.

The calculated RPM value is then sent to the 16×2 LCD display through the I2C module, where the speed is displayed in real time. At the same time, the LED indicator blinks whenever a pulse is detected, showing proper system operation.

Complete Working Steps

- 5V power supply turns ON the system
- DC motor rotates the fan blade
- IR sensor detects rotating fan movement
- Sensor generates digital pulse signals
- Pulse signals are sent to Arduino D9 pin
- Arduino counts pulses and calculates RPM
- RPM value is displayed on LCD
- LED blinks during pulse detection

Thus, the project accurately measures and displays the rotational speed of the motor using sensor detection, Arduino processing, and digital display technology.

CHAPTER 3

CIRCUIT DIAGRAM

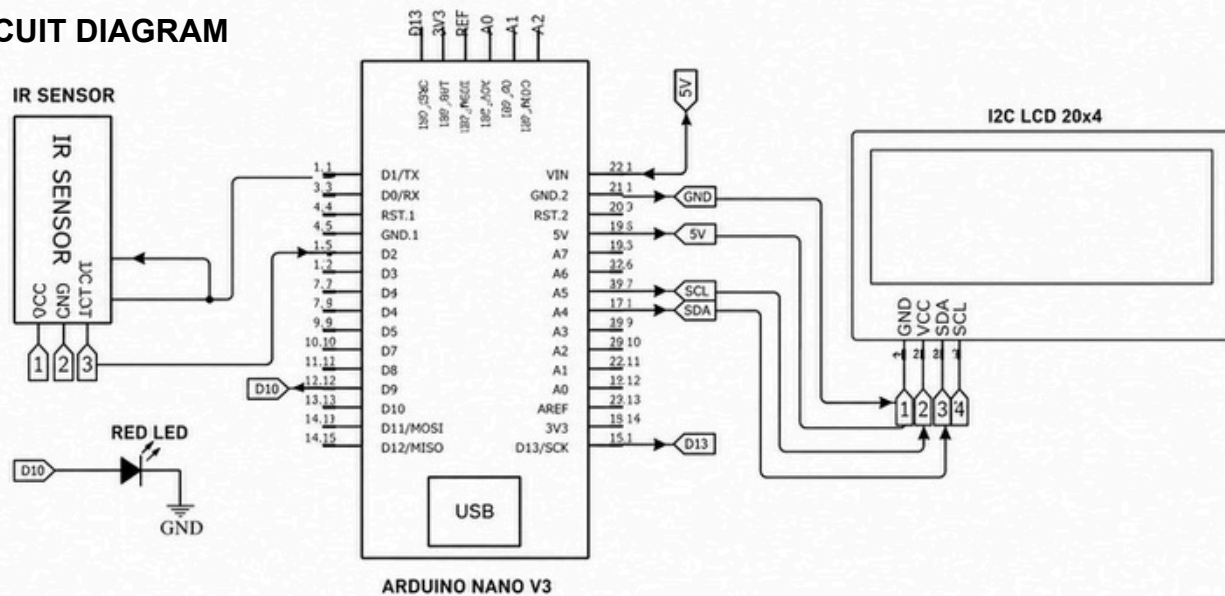


Fig 3.1

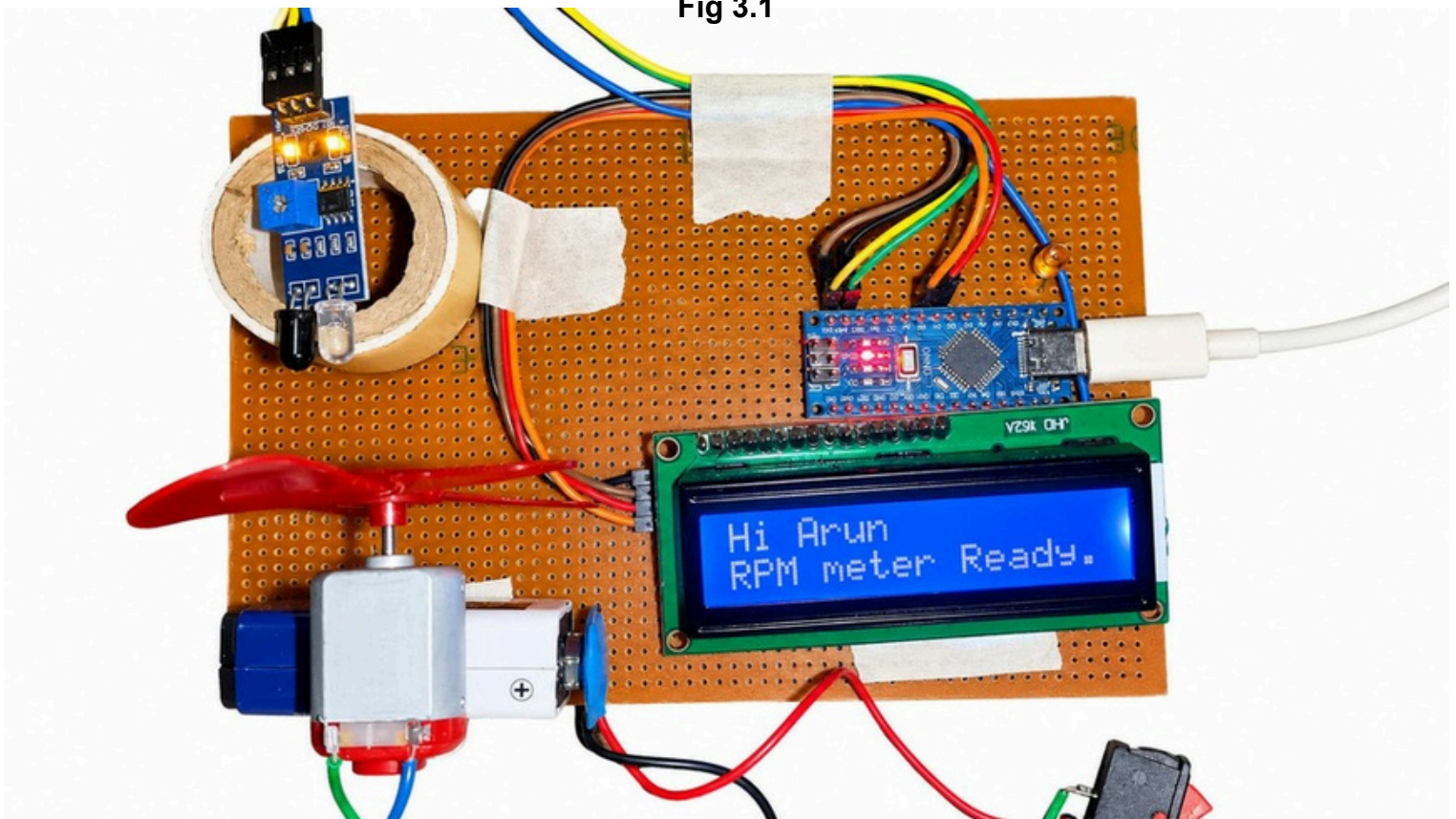


fig-3.2

Circuit Connections

- IR Sensor OUT → Arduino Digital Pin D2
- LCD SDA → Arduino A4
- LCD SCL → Arduino A5
- LCD VCC → 5V
- LCD GND → GND
- LED → Digital Pin through resistor

4.1 Mathematical Formula for RPM Calculation

In the Digital RPM Meter project, the rotational speed of the motor is calculated using pulse counting and time measurement techniques. The IR sensor generates one digital pulse whenever the reflective strip or fan blade passes in front of the sensor. These pulses are counted by the Arduino Nano within a fixed time interval.

If one pulse represents one complete rotation, then the RPM (Revolutions Per Minute) can be calculated by multiplying the number of pulses detected in one second by 60.

The mathematical formula used in this project is:

$$RPM = Pulses\ in\ 1\ Second \times 60$$

Example Calculation

If the IR sensor detects 25 pulses in one second:

$$RPM = 25 \times 60 = 1500$$

Therefore, the rotational speed of the motor is 1500 RPM.

Formula Explanation
Pulse = One complete rotation detected by IR sensor

1 Second = Fixed time interval used for counting

60 = Conversion factor from seconds to minutes

This formula allows the Arduino Nano to calculate motor speed accurately and display the RPM value in real time on the LCD screen.

4.2 Source Code Description

The Arduino program controls the complete operation of the Digital RPM Meter system. The code starts by including the required libraries for I2C communication and LCD display operation. The IR sensor input pin and LED output pin are defined in the program. The Arduino continuously reads the signal coming from the IR sensor connected to digital pin D9.

Whenever the rotating fan blade passes in front of the IR sensor, the sensor generates a HIGH pulse signal. The Arduino detects this signal transition and increases the pulse counter variable. At the same time, the LED connected to pin 13 blinks, indicating pulse detection.

The program uses the millis() timer function to measure a fixed time interval of one second. After every one second, the Arduino multiplies the total pulse count by 60 using the RPM formula to calculate the motor speed. The calculated RPM value is then displayed on the 16×2 LCD display through the I2C module. The LCD updates the RPM reading continuously in real time.

Arduino Source Code

#c++

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 16, 2);
#define IR_SENSOR 9
#define LED_PIN 13
volatile unsigned int pulseCount = 0;
unsigned long previousMillis = 0;
unsigned long displayMillis = 0;
int rpm = 0;
bool lastState = LOW;
void setup()
{
  pinMode(IR_SENSOR, INPUT);
  pinMode(LED_PIN, OUTPUT);
  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("Hi Arun");
  lcd.setCursor(0, 1);
  lcd.print("RPM meter Ready..");

  delay(2000);
  lcd.clear();
}
void loop()
{
  // Read sensor state
  bool currentState = digitalRead(IR_SENSOR);
  // Detect object only once
  if (currentState == HIGH && lastState == LOW)
  {
    pulseCount++;

    // LED blink
    digitalWrite(LED_PIN, HIGH);
    delay(10);
    digitalWrite(LED_PIN, LOW);
  }

  lastState = currentState;

  // Every 1 second calculate RPM
  if (millis() - previousMillis >= 1000)
  {
    previousMillis = millis();
```

```

/*
RPM Formula:
RPM = pulses in 1 sec × 60

If 1 rotation = 1 pulse
*/

rpm = pulseCount * 60;

pulseCount = 0;
}

// Refresh display every 200ms
if (millis() - displayMillis >= 600)
{
displayMillis = millis();

lcd.clear();

lcd.setCursor(0, 0);
lcd.print("Reading RPM");

lcd.setCursor(0, 1);
lcd.print("RPM: ");
lcd.print(rpm);
}
}

```

4.3 Power Supply Requirement

A stable 5V DC power supply is required for proper operation of the Digital RPM Meter system. The Arduino Nano, IR sensor module, 16×2 LCD display with I2C module, and LED indicator all operate on 5V supply.

A regulated USB supply or DC adapter is used to power the complete circuit. The regulated supply provides stable voltage, which helps in accurate sensor detection, proper Arduino processing, and clear LCD display operation.

Operating Voltage

Arduino Nano → 5V DC

IR Sensor → 5V DC

LCD with I2C → 5V DC

LED Indicator → 5V DC

Importance of Power Supply

- Provides stable system operation
- Prevents voltage fluctuations
- Improves RPM measurement accuracy
- Protects electronic components

Thus, a regulated 5V supply ensures reliable and accurate performance of the Digital RPM Meter system.

CHAPTER 5 – DESCRIPTION OF COMPONENTS

The Digital RPM Meter project consists of several electronic components that work together to measure and display the rotational speed of the motor accurately. Each component performs a specific function in signal detection, processing, calculation, and display operation.

The main components used in this project are **Arduino Nano, IR sensor module, 16×2 LCD display with I2C module, DC motor with fan, LED indicator, switch, battery, PCB, and connecting wires**. Proper selection and interfacing of these components ensure accurate RPM measurement, stable operation, and reliable performance of the system.

5.1 Arduino Nano

Arduino Nano is the main processing and controlling unit used in the Digital RPM Meter project. It is based on the ATmega328P microcontroller, which processes sensor signals, performs RPM calculations, and controls the LCD display. The Arduino Nano operates on 5V DC supply, which is provided through USB or regulated power input.

In this project, the Arduino receives pulse signals from the IR sensor, processes them using programmed coding, calculates RPM mathematically, and displays the output on the LCD screen in real time.

The Arduino Nano contains digital pins, analog pins, power pins, communication pins, and control pins that help in interfacing different electronic components.

Important Pins of Arduino Nano

- Digital Pins (D0–D13)

These pins are used for digital input and output operations.

- D9 Pin → Connected to OUT pin of IR sensor to receive pulse signals
- D13 Pin → Connected to LED indicator for pulse indication

Analog Pins (A0–A7)

These pins are mainly used for analog signal reading and communication purposes.

- A4 Pin (SDA) → Connected to SDA pin of I2C LCD module
- A5 Pin (SCL) → Connected to SCL pin of I2C LCD module

Power Pins

- 5V Pin → Supplies 5V power to IR sensor and LCD module
- GND Pin → Provides common ground connection for the complete circuit
- VIN Pin → Used for external voltage input to Arduino

Communication Pins

- TX Pin (D1) → Used for serial data transmission
- RX Pin (D0) → Used for serial data receiving

Reset Pin

RESET Pin → Used to restart or reset the Arduino program manually

Pin Connections Used in This Project

IR Sensor OUT → Arduino D9

LED Positive → Arduino D13

LCD SDA → Arduino A4

LCD SCL → Arduino A5

LCD VCC and Sensor VCC → Arduino 5V

LCD GND and Sensor GND → Arduino GND

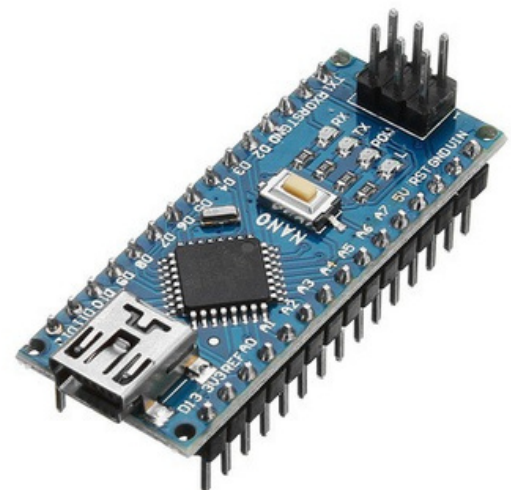


fig-5.1.1

The ATmega328P processor continuously reads pulse signals from the IR sensor through D9 pin. It counts these pulses, calculates RPM using coding and mathematical formulas, and sends the processed data to the LCD display through A4 and A5 communication pins.

Features of Arduino Nano

- Based on ATmega328P microcontroller
- Operates on 5V DC supply
- Compact and lightweight design
- 14 Digital I/O pins
- 8 Analog pins
- Easy programming using Arduino IDE
- Fast processing and reliable operation

Thus, Arduino Nano controls the complete Digital RPM Meter system by handling signal processing, RPM calculation, LED indication, and LCD display operation accurately and efficiently.

5.2 IR Sensor Module

The IR (Infrared) Sensor Module is used in the Digital RPM Meter project to detect the rotation of the motor fan and generate pulse signals for RPM calculation. It works on the principle of infrared light reflection and non-contact object detection.

The sensor module mainly contains an IR transmitter (IR LED), IR receiver (photodiode), comparator circuit, sensitivity adjustment potentiometer, and indicator LED. The IR transmitter continuously emits infrared rays toward the rotating fan blade or reflective strip. Whenever the reflective surface passes in front of the sensor, the infrared light reflects back to the receiver section.

The IR receiver detects the reflected light and the comparator circuit converts it into a digital output signal. This digital pulse signal is generated from the OUT pin of the sensor and sent to the Arduino Nano for processing.

The IR sensor module operates on 5V DC supply, which is provided from the 5V pin of the Arduino Nano.

Pins of IR Sensor Module

VCC Pin

- Connected to Arduino 5V pin
- Supplies power to the sensor module

GND Pin

- Connected to Arduino GND
- Provides common ground connection

OUT Pin

- Connected to Arduino D9 pin
- Sends digital pulse signals to Arduino

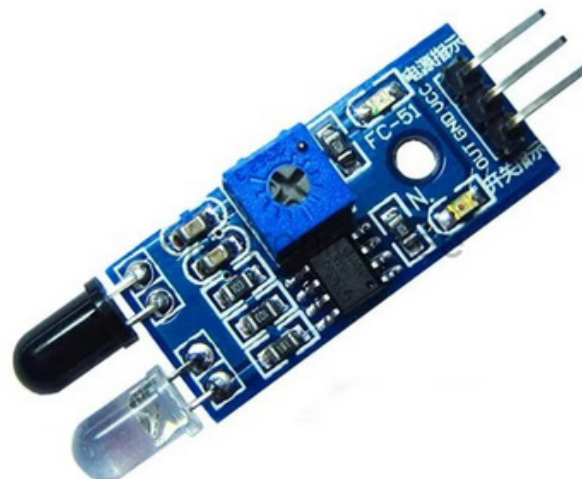
Working in This Project

- IR transmitter emits infrared light
- Reflective fan blade reflects IR rays
- IR receiver detects reflected light
- OUT pin generates pulse signal
- Arduino reads pulses from D9 pin
- RPM is calculated and displayed on LCD

Features of IR Sensor Module

- Operates on 5V DC supply
- Fast response time
- Non-contact speed detection
- Adjustable sensitivity control

fig-5. 2.1



- Accurate pulse generation
- Easy interfacing with Arduino

Thus, the IR sensor module plays an important role in detecting rotational movement and converting it into digital pulse signals for accurate RPM measurement.

5.3 16×2 LCD Display with I2C Module

A 16×2 LCD display is used in the Digital RPM Meter project to display the calculated RPM value in real time. The display can show 16 characters in 2 rows and provides a clear visual output for monitoring motor speed.

In this project, the LCD is connected to the Arduino Nano using an I2C module. The I2C module reduces wiring complexity and allows communication using only two wires. The LCD and I2C module operate on 5V DC supply, which is provided from the Arduino Nano.

After processing the pulse signals and calculating the RPM value, the Arduino sends the data to the LCD through the I2C communication lines. The LCD continuously updates and displays the RPM reading.

Pins of 16×2 LCD with I2C Module

VCC Pin

- Connected to Arduino 5V pin
- Provides power supply to LCD and I2C module

GND Pin

- Connected to Arduino GND
- Provides common ground connection

SDA Pin

- Connected to Arduino A4 pin
- Used for serial data communication

SCL Pin

- Connected to Arduino A5 pin
- Used for clock signal communication

Working in This Project

- Arduino calculates RPM value
- Data is sent through SDA and SCL lines
- I2C module receives the data
- LCD displays RPM value in real time

Features of LCD with I2C Module

- Operates on 5V DC supply
- Uses only two communication wires
- Reduces wiring complexity
- Displays RPM clearly in real time
- Easy interfacing with Arduino
- Low power consumption

Thus, the 16×2 LCD display with I2C module provides a simple, reliable, and user-friendly method for displaying RPM values accurately.

fig-5.3.1



5.4 DC Motor with Fan, Switch, and Battery

A DC motor with a fan is used in the Digital RPM Meter project as the rotating object whose speed is measured. The motor converts electrical energy into mechanical rotational motion, while the fan attached to the motor shaft provides visible rotation for RPM detection.

The DC motor operates on DC voltage supplied through a battery. A switch is connected in series with the motor to control the ON and OFF operation of the circuit. When the switch is turned ON, current flows from the battery to the motor, causing the fan blade to rotate.

The rotating fan blade passes in front of the IR sensor, which detects each rotation and generates pulse signals for RPM calculation.

Components and Connections

DCMotor

- Produces rotational movement
- Connected to battery supply through switch

Fan Blade

- Attached to motor shaft
- Used for rotation detection by IR sensor

Switch

- Controls power supply to motor
- Used for ON/OFF operation

Battery

- Provides DC power to motor circuit
- Supplies required voltage for motor rotation

Working in This Project

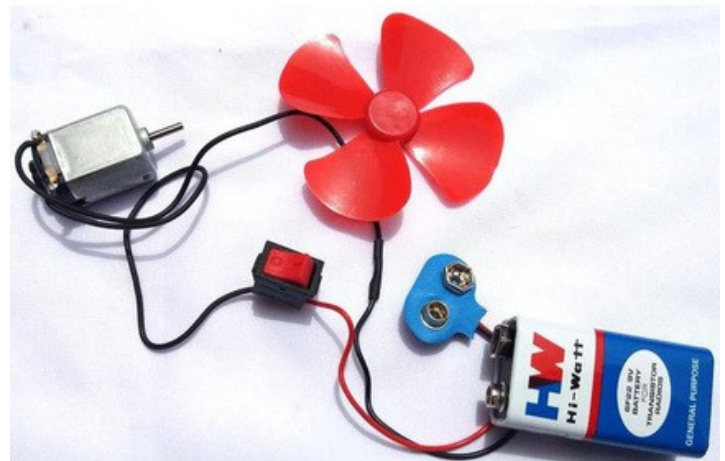
- Battery supplies DC power
- Switch turns motor ON/OFF
- Motor rotates the fan blade
- IR sensor detects fan rotation
- Pulse signals are generated for RPM measurement

Features

- Simple and compact setup
- Easy motor speed testing
- Portable power supply
- Smooth rotational operation
- Suitable for RPM measurement experiments

Thus, the DC motor, fan, switch, and battery together create the rotating mechanism required for accurate RPM detection and measurement in the project.

fig-5.4.1



5.5 Jumper Wires and PCB

Jumper wires and PCB (Printed Circuit Board) are used in the Digital RPM Meter project to provide proper electrical connections and organized circuit arrangement. They help in connecting all components such as Arduino Nano, IR sensor, LCD display, LED, motor, and power supply safely and efficiently.

Jumper wires are flexible insulated wires used for temporary and easy connections between components on the breadboard and Arduino Nano. Different colored wires are used for easy identification of power, ground, and signal connections.

The PCB or breadboard provides mechanical support and proper arrangement of the circuit components. It helps in reducing loose connections, improving circuit reliability, and making the project neat and compact.

Jumper Wires

- Used for electrical connections
- Connect Arduino with sensor and LCD
- Flexible and easy to use
- Different colors for easy identification

PCB / Breadboard

- Holds components properly
- Provides organized circuit layout
- Reduces connection errors
- Improves circuit reliability

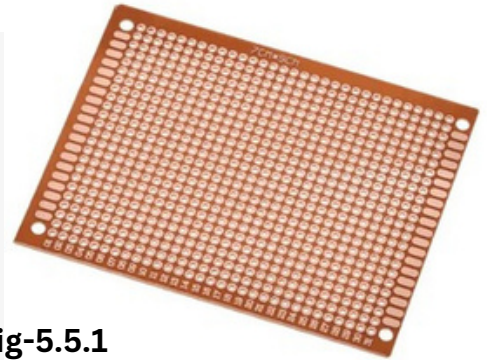


fig-5.5.1

Thus, jumper wires and PCB play an important role in building a reliable and properly connected Digital RPM Meter circuit.

5.6 Power Adapter for 5V Supply

A power adapter is used in the Digital RPM Meter project to provide a stable and regulated 5V DC supply to the Arduino Nano and other electronic components. Proper power supply is very important for accurate sensor detection, Arduino processing, and LCD operation.

In this project, the power adapter supplies voltage to the Arduino Nano through the USB port or VIN pin. The Arduino then provides 5V output to the IR sensor module and the 16×2 LCD display with I2C module through its 5V pin. A regulated 5V supply helps prevent voltage fluctuations and ensures stable operation of the complete system.

Role of Power Adapter

- Supplies regulated 5V DC power
- Powers the Arduino Nano
- Provides stable operation of sensor and LCD
- Prevents voltage fluctuation problems
- Improves overall system reliability

Connections in This Project

- Adapter output → Arduino Nano USB/VIN
- Arduino 5V pin → IR Sensor VCC
- Arduino 5V pin → LCD VCC
- Arduino GND → Common ground connection

Features

- Stable 5V DC output
- Easy power connection
- Reliable and safe operation
- Suitable for continuous working



fig-5.6.1

Thus, the power adapter ensures proper and reliable power distribution to all components of the Digital RPM Meter system.

CHAPTER 7 – ADVANTAGES

The Digital RPM Meter using Arduino offers many advantages due to its accurate sensing, real-time processing, compact design, and simple operation. The system is efficient, reliable, and suitable for various electronic and industrial applications.

Advantages of Digital RPM Meter

- **Accurate RPM Measurement**

The project measures motor speed accurately using IR sensor pulse detection and Arduino processing.

- **Real-Time Speed Monitoring**

RPM values are continuously updated and displayed instantly on the LCD screen.

- **Non-Contact Detection**

The IR sensor detects rotation without physical contact, reducing friction and mechanical wear.

- **Fast Response Time**

The system quickly detects pulses and calculates RPM within a short time interval.

- **Compact and Portable Design**

Arduino Nano and small electronic components make the project lightweight and easy to carry.

- **Low Power Consumption**

The circuit operates efficiently on 5V DC supply with minimum power usage.

- **Easy Circuit Design**

The use of I2C LCD reduces wiring complexity and makes the circuit simple and organized.

- **Easy Programming and Modification**

Arduino IDE provides simple coding, making future modifications and upgrades easier.

- **Reliable Performance**

The system provides stable operation and continuous RPM monitoring.

- **Cost-Effective System**

The project uses low-cost electronic components, making it affordable and practical.

- **Easy Troubleshooting**

LED indication and simple connections help in quick testing and fault detection.

- **Educational and Industrial Use**

The project is useful for laboratories, automation systems, robotics, motor testing, and learning embedded systems.

CHAPTER 8 – DISADVANTAGES

Although the Digital RPM Meter using Arduino provides accurate and reliable RPM measurement, the system also has some limitations and disadvantages during operation.

Disadvantages of Digital RPM Meter

Affected by External Light

- Strong sunlight or external infrared light may affect the performance of the IR sensor and reduce detection accuracy.

Dust and Dirt Sensitivity

- Dust particles on the sensor or reflective surface can interrupt IR reflection and generate incorrect readings.

Limited Detection Range

- The IR sensor works properly only within a short distance from the rotating object.

Proper Alignment Required

- Accurate RPM measurement depends on correct alignment between the IR sensor and reflective strip.

Not Suitable for Very High-Speed Motors

- At extremely high rotational speeds, pulse detection may become difficult without advanced sensors.

Dependence on Stable Power Supply

- Voltage fluctuations can affect Arduino processing, sensor performance, and LCD display operation.

Single Object Detection

- The system is mainly designed for one rotating object at a time and may require modification for multiple motor monitoring.

Limited Industrial Protection

- The basic circuit does not include protective casing against moisture, vibration, or harsh industrial environments.

CHAPTER 9 – APPLICATIONS

The Digital RPM Meter using Arduino is widely used in areas where accurate rotational speed measurement and monitoring are required. Due to its simple design, low cost, and reliable performance, the system can be used in educational, industrial, and automation applications.

Applications of Digital RPM Meter

Motor Speed Measurement

- Used to measure the rotational speed of DC motors and electric motors accurately.

Industrial Automation Systems

- Helps in monitoring and controlling motor speed in automated machines and industrial equipment.

Robotics Projects

- Used in robotic systems for wheel speed monitoring and motion control applications.

Fan Speed Monitoring

- Measures and displays the speed of cooling fans and exhaust fans in electronic systems.

Automotive Applications

- Used for RPM measurement in vehicle engines, motors, and testing systems.

Conveyor Belt Monitoring

- Helps in monitoring the speed of conveyor motors in industrial production systems.

Educational Laboratories

- Used in electronics and embedded system laboratories for practical learning and experimentation.

Research and Development

- Useful in testing and analysis of rotating machines and sensor-based projects.

Machine Maintenance Systems

- Helps in detecting abnormal motor speed and improves machine performance monitoring.

Embedded System Applications

- Demonstrates practical implementation of sensors, microcontrollers, and digital display systems.

CHAPTER 10 – CONCLUSION

The Digital RPM Meter using Arduino successfully demonstrates the measurement of rotational speed using sensor detection and microcontroller processing techniques. The project accurately measures the RPM of a DC motor fan using an IR sensor and displays the result on a 16×2 LCD display in real time.

In this system, the IR sensor detects the rotation of the fan blade and generates pulse signals, which are processed by the Arduino Nano. The Arduino counts the pulses, applies the RPM calculation formula, and displays the final RPM value on the LCD through the I2C communication module. The LED indicator also helps in showing pulse detection and proper system operation.

The project provides accurate speed measurement, fast response, low power consumption, simple circuit design, and reliable performance. The use of Arduino Nano makes the system easy to program, modify, and understand for embedded system applications.

This project also improves practical knowledge of Arduino programming, sensor interfacing, LCD communication, signal processing, and digital electronics. Due to its compact size, cost-effective design, and real-time monitoring capability, the Digital RPM Meter can be used in educational laboratories, industrial automation systems, robotics projects, motor testing, and research applications.

Thus, the Digital RPM Meter using Arduino is an efficient, user-friendly, and reliable system for accurate rotational speed measurement.

References

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- [Embedded Systems and Microcontroller Books](#)
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- [Basic Embedded C++ Programming References](#)